NATIONAL SPACE TEST AND TRAINING COMPLEX (NSTTC) INNOVATIVE TECHNOLOGY & ENGINEERING - SPACE TEST AND RANGE (NITE-STAR)

CAPABILITY DEVELOPMENT

MULTIPLE AWARD (MA) INDEFINITE DELIVERY INDEFINITE QUANTITY (IDIQ)

STATEMENT OF WORK (SOW)

UNITED STATES SPACE FORCE (USSF)

SSC/TI

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1. General Information - Background

1.1. Mission Overview

The Operational Test and Training Infrastructure (OTTI) Integrated Program Office (IPO) mission is to provide space warfighters interconnected, scalable, and distributed live/physical and synthetic/digital ranges for full-spectrum test and training. This infrastructure enables the development, validation, and sharpening of joint warfighting solutions to prevail in contested and operationally challenging space environments.

1.2. Purpose

The NSTTC Innovative Technology & Engineering – Space Test and Range (NITE-STAR) Capability Development Multiple Award (MA) Indefinite-Delivery, Indefinite-Quantity (IDIQ) base contract vehicle will be used to support U.S. Space Force (USSF) OTTI by designing, developing, integrating, testing, and sustaining advanced space-based and ground-based systems. These systems include, but are not limited to:

1.2.1 Space Based Test and Training Systems such as satellite vehicles, spacecraft subsystems, and payloads.

1.2.2 Ground-Based Test and Training Systems such as ground stations, ground sensors, instrumentation, antennas, telescopes, radars, command and control (C2) systems, and supporting infrastructure.

1.3. Ordering Period

The NITE-STAR Capability Development MA IDIQ has a five (5) year ordering period with one (1) five (5) year optional ordering period. The overall period-of-performance (PoP) for Task Orders (TOs) or Delivery Orders (DOs) can extend beyond the ordering period, but performance shall not extend more than one (1) year beyond the optional ordering period end date.

2. Scope

The NITE-STAR Capability Development MA IDIQ base contract will support space-based and ground-based capabilities development activities that shall enhance test, evaluation, and training operations across live, virtual, and constructive (LVC) environments, ensuring systems are tested and Guardians are prepared for operationally realistic space-based engagements and advanced threat scenarios. TOs/DOs awarded under this contract are intended to support the objectives of the OTTI Program Executive Officer (PEO) and support will extend to other stakeholders within the space test and training enterprise. The OTTI IPO will issue individual TOs/DOs in one or more areas listed below:

- Space Based and/or Ground Based Systems Development
- Space Based and/or Ground Based Systems Sustainment
- Enterprise and Range Enhancements
- Special Studies

3. Contractor Tasking/Requirements

The contractor shall perform tasks in accordance with separately issued TOs or DOs as specified therein.

3.1 Program Management

The contractor shall manage all aspects of the NITE-STAR Capability Development MA IDIQ contract and awarded TOs/DOs, to include cost, schedule, performance, and contract management.

3.1.1. The contractor shall act as the task integrator, ensuring all technical and program elements of a TO/DO provide a fully integrated approach.

3.1.2. The contractor shall provide status on the NITE-STAR Capability Development MA IDIQ TOs/DOs, to include program management, schedule, risk management, accomplishments, issues, and upcoming events.

3.1.3. The contractor shall provide notice to the Government Program Manager and Contracting Officer of who their designated Technical Leads are for individual TOs/DOs. The Technical Lead shall serve as the primary point of contact for all technical aspects of the required work.

3.1.4. The contractor shall provide Contract Data Requirements Lists (CDRLs) associated with respective TOs/DOs.

3.1.5. The contractor and its employees shall not supervise, direct, or control the activities of Government personnel or the employees of any other contractor, other than those subcontracted under the prime.

3.2 Systems Engineering

The contractor shall perform system engineering functions to include, but not limited to, requirements management, development, integration, verification, and sustainment to ensure all systems, subsystems, interfaces, and components are engineered to meet Government requirements, comply with DoD and USSF standards, and sustain mission effectiveness of capabilities as required at the TO/DO level.

3.3 Space and Ground Systems Development

The contractor shall develop, manufacture, integrate, deliver, and deploy Space and Ground systems in support of test, training, and range environments.

3.3.1 The contractor shall design, build, and deliver spacecraft vehicles and onboard space payloads to include but not limited to sensors and communications relay.

3.3.2 The contractor shall develop, integrate, and deploy ground stations, sensors, tracking, and Command and Control (C2) architecture.

3.3.3 The contractor shall design and develop Hardware-in-the-Loop (HWIL) and Software-in-the-Loop (SWIL) to simulate space and/or ground systems or operational environments.

3.3.4 The contractor shall develop digital twins that create real-time, dynamic system representations for validation, testing, predictive performance analysis, optimization as well as adaptive training environments for OTTI and NSTTC space and ground system operators.

3.3.5 The contractor shall integrate, leverage, and deploy Government furnished property/information (GFP/GFI) to support Space and Ground Systems activities outlined in 3.3.1 through 3.3.4. This includes the associated network(s), cryptography, data transport, data storage, data archive, and software development in accordance with the OTTI data strategy to the maximum extent feasible as determined by the Government.

3.4 Space Based and/or Ground Based Systems Sustainment

The contractor shall deliver comprehensive sustainment and lifecycle management for space and ground systems, ensuring long-term OTTI use, cybersecurity compliance, configuration control, and performance optimization.

3.5 Enterprise and Range Enhancements

The contractor shall provide enterprise-level and range capability enhancements to support space and ground, test, training, and operational readiness objectives. The contractor shall focus on improving range infrastructure, automation, interoperability, cybersecurity, and digital transformation to ensure test and training environments remain mission ready.

3.6 Special Studies

The contractor shall conduct special studies, technical analyses, and mission driven assessments of live and/or digital space and ground system development, infrastructure, integration, testing, and sustainment. Studies shall focus on risk identification, technology assessments, and feasibility evaluations that ensure a forward looking, data-driven approach to enhancing OTTI mission capabilities.

4. Guidance and Best Practices

The contractor should leverage industry best practices and open standards to develop and deliver necessary capabilities at scale and speed such as:

4.1 Utilizing digital engineering tools and Model Based Systems Engineering (MBSE) to develop end-to-end systems architectures and systems models to capture system behavior, structure, define/validate interface control, and full traceability from Government requirements, contractor's design, to verification artifacts to ensure continuous system lifecycle management from development to decommissioning. The contractor should share interactive digital engineering dashboards allowing real time access to systems architectures, models, dependencies, and status with the Government.

4.2 Following Development, Security, and Operations (DevSecOps) principles, Modular Open Systems Architectures (MOSA), automation, and an agile software development process that streamline development, manage dependencies/risks, increase velocity, and provide transparency across all development efforts for space and ground assets of the test and training enterprise systems of systems architecture.

4.3 Employing User Interface/User Experience (UI/UX) design, guides, and best practices.

4.4 Utilizing Government-approved requirements management tool to store, track, and update all system requirements.

4.5 Utilizing cybersecurity, Defensive Cyber Operations (DCO) and multi-level security and continuous Authority to Operate (cATO)/continuous Risk Management Framework (cRMF) approaches.

5. Security

The contractor shall implement, administer, and maintain a security program to ensure classified information is protected per the NITE-STAR Capability Development MA IDIQ contract DD Form 254, Department of Defense Contract Security Classification Specification, and TO/DO-specific appropriate Security Classification Guide(s).

5.1. The contractor shall have access to a Secure Compartmented Information Facility (SCIF), with Special Access Program-Facility (SAP-F) overlay that has been accredited for performing and safeguarding work up to Top Secret/Sensitive Compartmented Information (TS/SCI) and Special Access Program (SAR/SAP) at time of contract award.

5.1.1. The contractor may be excluded from a TO/DO award if unable to maintain facility requirements.

5.1.2. The contractor shall work with the Government Program Security Officer (PSO) direction to ensure security issues are properly addressed and Government approval is obtained when necessary.

5.2. The contractor shall ensure all assigned personnel are aware that any time they communicate to or from a Government facility, they are subject to COMSEC procedures. All communications within DoD organizations are subject to COMSEC review. The Government may conduct COMSEC monitoring and recording of electronic or voice communications originating from or terminating at DoD organizations at any time.

5.3. The contractor shall maintain personnel clearances and System Administration/Cybersecurity (previously Information Assurance) DoD 8570 certifications commensurate with the DD Form 254.

5.4. The contractor shall provide contractor support to administer SCI clearances, provide security resources (or support) to perform TOs/DOs work at the SCI level, and coordinate with the Government for sufficient and necessary number of SCI billets.

5.5. The contractor shall comply with all Defense Security Service (DSS) security requirements for computing systems and networks.

5.6. The contractor shall ensure wide and frequent dissemination of the information to all employees dealing with Government information.

5.7. The contractor shall apply NIST SP 800-37 and NIST SP 800-53 to implement Cybersecurity and Risk Management Framework (RMF).

5.8. The contractor shall perform system and cyber security IAW current regulations and contract requirements including, but not limited to, DoDI 8500.01, DoDI 8510.01, DoDI 8530.01, DoD 8750.01-M.

5.9. The contractor shall access contractor facilities, Government facilities, documents, and systems in accordance with the NITE-STAR Capability Development MA IDIQ DD Form 254.

5.10. The contractor shall provide initial notifications and final security incident reports for all reportable security incidents involving OTTI contracts, personnel, etc. to OTTI security for tracking/oversight purposes.

5.11. The contractor shall provide their annual inspection results (whether conducted by DCSA, OSI PJ, or other Cognizant Security Authorities) to OTTI security for tracking/oversight purposes.

6. Travel / Temporary Duty (TDY)

The Contractor shall travel as required in support of the individual TOs/DOs in accordance with the Joint Travel Regulation (JTR) in effect at the time of travel. The need for travel will be identified at the order level.

7. Materials & Government Furnished Property/Information

If a TO/DO requires specific materials, new software tools, or hardware, the materials will be identified at the order level. The contractor shall procure these materials according to the Government approved Bill of Materials (BOM) in the specific TO/DO. The contractor shall procure these materials in accordance with their approved procurement system and disclosure statement.

8. Government Furnished Property/Information (GFP/GFI)

Any GFP/GFI required to execute work under this IDIQ will be addressed at the TO/DO level.

9. Data Rights

Applicable Technical Data, Computer Software, and Computer Software Documentation data rights will be addressed at the TO/DO level.

10. Quality Assurance

The contractor shall implement established quality assurance processes and procedures and further define additional quality assurance processes and procedures as needed to ensure the reliability, durability, and long-term performance of system components.

11. Training and Support

11.1. At the TO/DO level, the contractor shall provide training, training materials and CONOPS development support to range operators, range maintenance personnel, end-users, and stakeholders for systems developed under NITE-STAR Capability Development MA IDIQ.

11.2. At the TO/DO level, the contractor shall provide technical support for space and ground systems developed under NITE-STAR Capability Development MA IDIQ including documentation, user guides, and any additional resources available to assist users. The specific time frame for technical support will be defined at the TO/DO level.

12. Documentation

12.1. The contractor shall create comprehensive documentation, including ground and space system designs, specifications, user manuals, installation guides, test plans, operational procedures, and technical reports. Documentation should also include space and ground instrumentation descriptions, digital model descriptions, model-based systems engineering processes, operator training materials and, where applicable, training systems/hardware guides. All required documentation will be delivered based on the established CDRLs on the individual TO/DO.

12.2. The contractor shall provide comprehensive lifecycle management and logistics support plans to ensure optimal operations, maintenance, and sustainability of systems developed. All required plans will be delivered based on the established CDRLs on the individual the TO/DO.

13. Associate Contractor Agreement (ACA)

13.1. At the TO/DO level the contractor shall enter into Associate Contractor Agreements (ACA) for any portion of the contract requiring joint participation in the accomplishment of the Government's requirement. The agreements shall include the basis for sharing information, data, technical knowledge, expertise, and/or resources essential to the NITE-STAR Capability Development MA IDIQ that shall ensure the greatest degree of cooperation for program development to meet the terms of the contract.

13.2. ACAs shall include the following general information:

- Identify the associate contractors and their relationships.
- Identify involved program and the relevant associate contractors Government contracts.
- Describe the associate contractor interfaces by general subject matter.
- Specify the information categories to be exchanged or support to be provided.
- ACA expiration date.
- Identify potential conflicts between relevant Government contracts and the ACA; include agreements on employee proprietary data protection and restrictions.

14. Government Team

14.1. The Government team includes Non-Government Advisors (NGAs) as follows: Federally Funded Research and Development Corporations (FFRDCs), University-Affiliated Research Centers (UARCs), and Government contractors. NGAs are subject to change during the performance of the NITE-STAR Capability Development MA IDIQ and the list identified in paragraph 13.2 may be updated unilaterally by the Government as a result.

14.2. The contractor shall notify the Contracting Officer(s) immediately if there are any concerns regarding the participation of NGA personnel during the performance of the contract.

14.2.1. FFRDCs: The Aerospace Corporation, MITRE Corporation, and Massachusetts Institute of Technology-Lincoln Laboratory (MIT-LL),

14.2.2. UARCS: Johns Hopkins University Applied Physics Laboratory (JHU/APL), Utah State University Space Dynamics Laboratory (SDL), and Lawrence Livermore National Laboratory (LLNL), and Georgia Tech Research Institute (GTRI).

14.2.3. Advisory and Assistance Services (A&AS) contractors: Galapagos Federal Systems LLC, Science Applications International Corporation (SAIC), General Dynamics Information Technology (GDIT), ManTech International Corporation, Tyto Athene LLC, ENSCO, KBR, and Tecolote Research, Inc.

ANNEX A: ENABLING REQUIREMENTS FOR SSC PROGRAM CONTRACTS REQUIRING INTERFACE WITH AEROSPACE FFRDC CONTRACT SUPPORT

Overview. This contract covers part of a program which is under the general program management of the United States Space Force Space Systems Command (SSC). SSC has entered into a contract with The Aerospace Corporation, a California nonprofit corporation operating a Federally Funded Research and Development Center (FFRDC), for the services of a technical group that will support the program office by performing General Systems Engineering and Integration (GSE&I), Technical Review (TR), and/or Technical Support (TS), and informing the commander or director of the relevant organizations and programs it supports of product or process defects and other relevant information.

GGSE&I. This function involves overall system definition; integration both within the system and with associated systems; analysis of system segment and subsystem design; design compromises and tradeoffs; definition of interfaces; review of hardware and software, including manufacturing and quality control; observation, review and evaluation of tests and test data; support of launch, flight test, and orbital operations; assessment of the contractors' technical performance through meetings with contractors and subcontractors, exchange and analysis of information on progress and problems; review of plans for future work; developing solutions to problems; technical alternatives for reduced program risk; providing written comments and recommendations to the applicable Program Manager and/or Project Officer as an independent technical assessment for consideration for modifying the program or redirecting the contractor's efforts; all to the extent necessary to assure timely and economical accomplishment of program objectives consistent with mission requirements.

TTR. This function includes the process of appraising the technical performance of the contractor through meetings, exchanging information on progress and problems, reviewing reports, evaluating presentations, reviewing hardware and software, witnessing and evaluating tests, analyzing plans for future work, evaluating efforts relative to contract technical objectives, and providing comments and recommendations in writing to the applicable Program Manager and/or Project Officer as an independent technical assessment for consideration for modifying the program or redirecting the contractor’s efforts to assure timely and economical accomplishment of program objectives.

TTS. This function involves broad areas of specialized needs of customers for planning, system architecting, research and development, horizontal engineering, or analytical activities for which The Aerospace Corporation is uniquely qualified by virtue of its specially qualified personnel, facilities, or corporate memory. The categories of TS tasks are: Selected Research, Development, Test and Evaluation; Plans and System Architecture; Multi-Program Systems Enhancement; International Technology Assessment; and Acquisition Support.

Prime contractor requirement. In the performance of this contract, the contractor agrees to cooperate with The Aerospace Corporation by: 1) responding to invitations from authorized U. S. Government personnel to attend meetings; 2) providing access to technical information and research, development planning data such as, but not limited to, design and development analyses, test data and results, equipment and process specifications, test and test equipment specifications and procedures, parts and quality control procedures, records and data, manufacturing and assembly procedures, and schedule and milestone data, all in their original form or reproduced form and including top-level life cycle cost* data, where available; 3) delivering data as specified in the Contract Data Requirements List; 4) discussing technical matters relating to this program; 5) providing access to contractor facilities utilized in the performance of this contract; 6) and allowing observation of technical activities by appropriate technical personnel of The Aerospace Corporation. The Aerospace Corporation personnel engaged in GSE&I, TR, and/or TS efforts: (i) are authorized access to all such technical information (including proprietary information) pertaining to this contract and may discuss and disclose it to applicable personnel in a program office; (ii) are authorized to discuss and disclose such technical information (including proprietary information) to the commander or director of the relevant organizations and programs; and (iii) Aerospace shall make the technical information (including proprietary information) available only to its Trustees, officers, employees, contract labor, consultants, and attorneys who have a need to know.

Subcontractor Requirement. The contractor further agrees to include in all subcontracts a requirement requiring compliance by subcontractor and supplier and succeeding levels of subcontractors and suppliers with the response and access and disclosure provisions of this Enabling Requirement, subject to coordination with the contractor, except for subcontracts for commercial items or commercial services. This agreement does not relieve the contractor of its responsibility to manage the subcontracts effectively and efficiently nor is it intended to establish privity of contract between the Government or The Aerospace Corporation and such subcontractors or suppliers, except as indicated below.

Master Non-disclosure Agreement. The Aerospace Corporation shall protect the proprietary information of contractors, subcontractors, and suppliers in accordance with its Master Non-disclosure Agreement, a copy of which is available upon request. This Master Non-disclosure Agreement satisfies the Nondisclosure Agreement requirements set forth in 10 U.S.C. §2320 (f)(2)(B), and provides that such contractors, subcontractors, and suppliers are intended third-party beneficiaries under the Master Non-disclosure Agreement and shall have the full rights to enforce the terms and conditions of the Master Non-disclosure Agreement directly against The Aerospace Corporation, as if they had been signatory party hereto. Each such contractor, subcontractor, or supplier hereby waives any requirement for The Aerospace Corporation to enter into any separate company-to-company confidentiality or other non-disclosure agreements.

Aerospace shall make the technical information (including proprietary information) available only to its Trustees, officers, employees, contract labor, consultants, and attorneys who have a need to know, and Aerospace shall maintain between itself and the foregoing binding agreements of general application as may be necessary to fulfill their obligations under the Master Non-disclosure Agreement referred to herein, and Aerospace agrees that it will inform contractors, subcontractors, and suppliers if it plans to use consultants, or contract labor personnel and, upon the request of such contractor, subcontractor, or supplier, to have its consultants and contract labor personnel execute non-disclosure agreements directly therewith.

Technical Direction. The Aerospace Corporation personnel are not authorized to direct the contractor in any manner. Technical direction under this contract will be provided to the contractor solely by SSC.

*"Cost data" means information associated with the programmatic elements of life cycle (concept, development, production, operations, and retirement) of the system/program. As defined, cost data differs from "financial" data, which is defined as information associated with the internal workings of a company or contractor that is not specific to a project or program."